

Project Title Here

Scalable Data Engineering Solution

Data Engineering I (1TD169) – Spring 2026

Uppsala University
Department of Information Technology

Team Members: Group 32

Abdur Rehman Khalid
Arnab Kumar Ghosh
Dip Chowdhury
Muhammad Umair
Pradip Kumar Das
Zihao Yang

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1 Background

[Describe the scientific/application context of dataset. Explain the domain, why this dataset is important or interesting, what analyses others have performed with it (cite 2–3 relevant papers/resources), and state specific analysis objective.]

2 Data Format & Characteristics

[Describe the dataset technically: format (CSV, JSON, Parquet, etc.), schema/structure, total size and number of records, data quality issues encountered, and justification for format choice or conversion.]

3 System Architecture & Implementation

[This is the main section. Include a system design diagram, justify choice of tools, describe the infrastructure configuration, explain the data pipeline stages, present code structure and key algorithms, and discuss challenges faced and solutions.]

3.1 System Design

[Include a system architecture diagram (mandatory). Describe the high-level design of distributed processing solution.]

3.2 Technology Stack

[Justify choice of tools: Spark vs. Hadoop vs. hybrid. Explain why the selected stack is suitable for workload.]

3.3 Infrastructure Configuration

[Describe the VM setup and cluster topology. Specify the number of VMs, flavors (vCPU, RAM), storage allocation, and network configuration.]

3.4 Data Pipeline

[Describe the data pipeline stages: ingestion → processing → output. Explain the ETL workflow and key transformations applied.]

3.5 Code Structure

[Describe the repository layout and key modules (etl_job.py, analysis_job.py, config.py, etc.). Explain the role of each component.]

3.6 Challenges & Solutions

[Discuss the main engineering challenges encountered during the project and how you resolved them.]

4 Scalability Experiments & Results

[Present experimental findings: experimental design, results with plots, analysis of results, bottleneck identification, and comparison to theoretical expectations.]

4.1 Experimental Design

[Describe what you varied in experiments (number of workers, cores, data size), the metrics measured (runtime, throughput), and the methodology used.]

4.2 Horizontal Scaling Results

[Present results of horizontal (strong) scaling experiments: fixed dataset size with increasing number of worker nodes. Include runtime vs. number of workers plot.]

4.3 Vertical Scaling Results

[Present results of vertical scaling experiments: fixed number of nodes with increasing cores/memory per node. Include runtime vs. resources plot.]

4.4 Speedup & Efficiency Analysis

[Compute and plot the speedup factor ($S = T_1/T_N$) and efficiency ($E = S/N$). Compare against ideal linear speedup. Discuss deviations.]

4.5 Bottleneck Identification

[Identify what limits scalability: CPU, network I/O, disk I/O, shuffle overhead, data skew, or serialization. Reference Amdahl's Law if applicable.]

5 Discussion & Conclusion

[Reflect on the project: Was the architecture suitable? What worked well? What would you improve with more time/resources? What did you learn about distributed systems?]